

1. HUB

☐ বাংলা Concept

Hub হলো একটি **Basic Networking Device** যা একাধিক Computer-কে সংযুক্ত করে। এটি কোনো Data Check করে না; বরং একটি Port-এ Data পেলে সব Port-এ পাঠিয়ে দেয় (Broadcast)।

✓ Exam Ready Answer (English)

A **Hub** is a basic networking device that connects multiple computers in a LAN. It broadcasts incoming data to all connected devices without checking the destination address.

Easy Example

Suppose PC-1 sends data to PC-2. A **Hub** sends the data to **PC-2, PC-3, and PC-4**. Only PC-2 accepts it; others ignore it.

Key Points

- Layer: **Physical Layer (Layer 1)**
 - Intelligent Device: **✗ No**
 - Speed: Slow
 - Data Transmission: Broadcast
-

2. SWITCH ★ ★ ★ (Most Important)

☐ বাংলা Concept

Switch-ও একাধিক Computer-কে সংযুক্ত করে, কিন্তু এটি **MAC Address** দেখে শুধুমাত্র সঠিক Device-এ Data পাঠায়।

✓ Exam Ready Answer (English)

A **Switch** is a networking device that connects multiple devices in a LAN and forwards data only to the intended destination using **MAC addresses**.

Easy Example

Suppose PC-1 sends data to PC-2.

The **Switch** checks the MAC address and sends the data **only to PC-2**.

Key Points

- Layer: **Data Link Layer (Layer 2)**
- Intelligent Device: ✓ Yes
- Speed: Fast
- Data Transmission: Unicast

Used In

Banks, Offices, Universities

3. ROUTER ★ ★ ★

□ বাংলা Concept

Router দুটি বা ততোধিক **Different Network**-কে সংযুক্ত করে এবং **IP Address** ব্যবহার করে Data-এর সঠিক পথ (Route) নির্ধারণ করে।

✓ Exam Ready Answer (English)

A **Router** is a networking device that connects different networks and forwards data packets based on **IP addresses**.

Easy Example

Your home Wi-Fi Router connects your **Home Network** to the **Internet**.

Key Points

- Layer: **Network Layer (Layer 3)**
- Uses IP Address

- Connects Different Networks

4. BRIDGE

□ বাংলা Concept

Bridge একই ধরনের দুটি LAN Segment-কে সংযুক্ত করে এবং অপয়োজনীয় Traffic কমায়।

✓ Exam Ready Answer

A Bridge is a networking device that connects two similar LAN segments and filters traffic using MAC addresses.

Easy Example

Two office LANs connected through a Bridge.

Key Points

- Layer: Data Link Layer (Layer 2)
- Filters Traffic

5. REPEATER

□ বাংলা Concept

দূরে Data পাঠাতে গেলে Signal দুর্বল হয়ে যায়।

Repeater সেই দুর্বল Signal-কে আবার শক্তিশালী (Regenerate) করে।

✓ Exam Ready Answer (English)

A Repeater is a networking device that regenerates weak signals to extend the transmission distance.

Easy Example

Long Optical Fiber Cable

Key Points

- Layer: Physical Layer
- Regenerates Signal

6. MODEM

□ বাংলা Concept

Modem Digital Signal-কে Analog এবং Analog Signal-কে Digital-এ রূপান্তর করে।

(Modulator + Demodulator)

✓ Exam Ready Answer (English)

A Modem is a device that converts digital signals into analog signals and analog signals into digital signals for Internet communication.

Easy Example

Broadband Internet Connection

7. GATEWAY

□ বাংলা Concept

ভিন্ন ধরনের Network বা Protocol-এর মধ্যে যোগাযোগ করায়।

✓ Exam Ready Answer (English)

A Gateway is a networking device that connects two different networks using different communication protocols.

Easy Example

A Bank System communicating with a Government Server using different protocols.

8. ACCESS POINT (AP)

□ বাংলা Concept

Wireless Device-কে LAN-এর সাথে সংযুক্ত করে।

✓ Exam Ready Answer (English)

An Access Point (AP) is a networking device that allows wireless devices to connect to a wired network.

Easy Example

Office Wi-Fi

9. FIREWALL ★ ★ ★

□ বাংলা Concept

Firewall Network-এর Security Guard-এর মতো কাজ করে।

এটি ক্ষতিকর Traffic Block করে এবং অনুমোদিত Traffic Allow করে।

Network

□ বাংলা Concept

Network হলো দুই বা ততোধিক Computer, Server, Printer, Router বা অন্যান্য Device-কে Cable অথবা Wireless-এর মাধ্যমে সংযুক্ত করার ব্যবস্থা, যাতে তারা **Data, Information এবং Resources Share** করতে পারে।

✓ Exam Ready Answer (English)

A Computer Network is a collection of two or more interconnected computers or devices that communicate and share data, resources, and services through communication links.

Easy Example

Suppose all computers of **Pubali Bank** are connected so that every branch can access customer accounts and process transactions securely. This is called a **Computer Network**.

Network Topology

□ বাংলা Concept

Network Topology হলো একটি Network-এর সকল Device (Computer, Switch, Hub, Router, Server)-এর **Physical অথবা Logical Arrangement**।

সহজভাবে,

Network = Devices Connected

Topology = Devices কীভাবে Connected আছে।

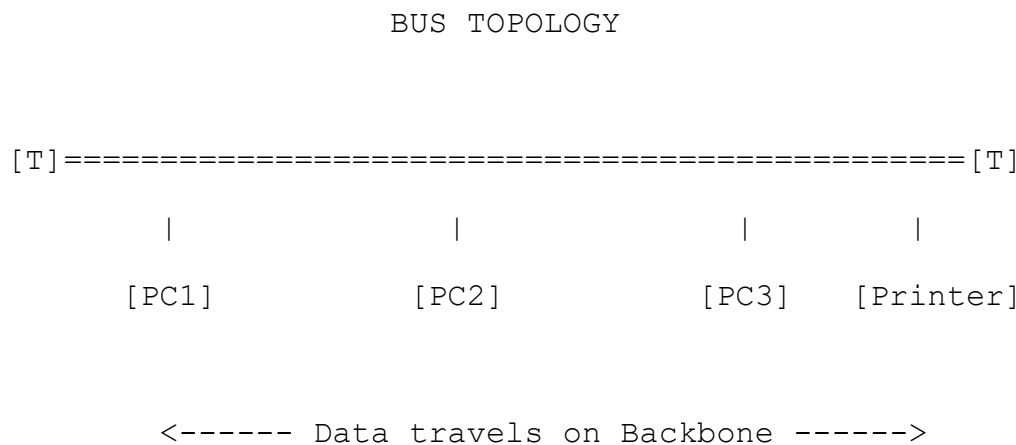
✓ Exam Ready Answer (English)

Network Topology is the physical or logical arrangement of computers, networking devices, and communication links in a computer network.

It defines how devices are connected and how data is transmitted between them.

1. BUS TOPOLOGY

Figure



T = Terminator

Bus Topology

□ বাংলা Concept

Bus Topology হলো এমন একটি Network Topology যেখানে সব Computer বা Device একটি মাত্র Main Communication Cable (Backbone Cable)-এর সাথে সংযুক্ত থাকে।

যখন কোনো Computer Data পাঠায়, তখন Data Backbone Cable-এর মাধ্যমে সব Computer-এর কাছে Broadcast হয়। এরপর প্রত্যেক Computer Destination MAC Address পরীক্ষা করে। যার MAC Address মিলে, শুধুমাত্র সেই Computer Data গ্রহণ (Accept) করে; অন্যরা Data উপেক্ষা (Discard) করে।

Backbone Cable-এর দুই প্রান্তে **Terminator** থাকে, যা Signal-এর প্রতিফলন (Signal Reflection) প্রতিরোধ করে।

✓ Exam Ready Answer (English)

Bus Topology is a network topology in which all computers or devices are connected to a single communication cable called the **Backbone Cable**.

When a device sends data, it is **broadcast** through the backbone cable to all connected devices. Each device checks the **destination MAC address** in the data frame. Only the device whose MAC address matches the destination accepts the data, while all other devices discard it.

A **Terminator** is placed at both ends of the backbone cable to absorb signals and prevent **signal reflection**, ensuring reliable communication.

Advantages

- Low Cost
- Easy Installation
- Less Cable Required

Disadvantages

- Backbone Failure → Entire Network Stops

Example

Small Office Network

[T]===== [T]

Backbone Cable (Main Bus Cable)

| | | |

[PC-1] =====> [PC-2] =====> [PC-3] =====> [PC-4]

Ignore Accept Ignore

2. STAR TOPOLOGY ★ ★ ★ (Most Star Topology-তে সব Computer একটি Central Switch/Hub-এর সাথে সংযুক্ত থাকে।

যখন একটি Computer অন্য Computer-কে Data পাঠায়, তখন Data প্রথমে Switch-এর কাছে যায়। Switch Destination MAC Address দেখে শুধুমাত্র নির্দিষ্ট Computer-এর কাছে Data পাঠায়।

এজন্য অপ্রয়োজনীয় Computer Data পায় না এবং Network দ্রুত (Fast) ও নিরাপদ (Efficient) হয়।

✓ Exam Ready Answer (English)

Star Topology is a network topology in which all devices are connected to a central **Switch** or **Hub**.

When a device sends data, it first reaches the central switch. The switch checks the **destination MAC address** and forwards the data only to the intended destination device.

Figure

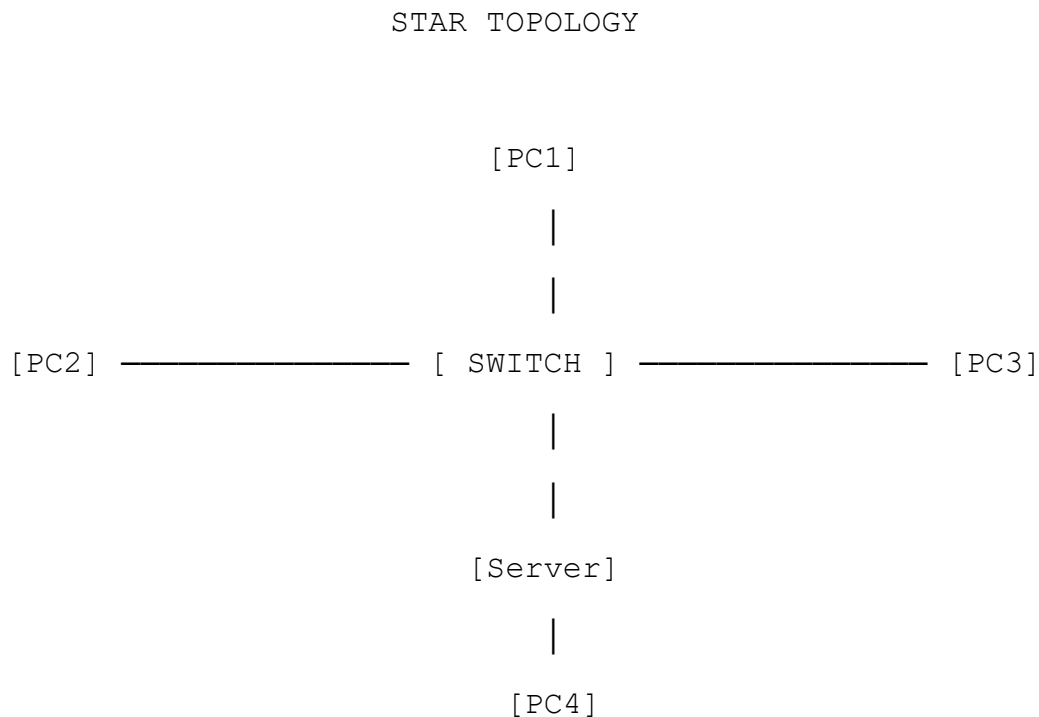


Figure Explanation

- **Switch** → Central Device
- **PCs** → Connected Devices
- All communication passes through the Switch.

Advantages

- Easy Management
- High Performance
- Easy Troubleshooting
- Easy Expansion

Disadvantages

- Switch Failure → Entire Network Stops

Example

Banks, Offices, Universities

3. RING TOPOLOGY

Ring Topology-তে Data একটি নির্দিষ্ট দিক (Clockwise বা Counterclockwise) অনুসরণ করে একটি Computer থেকে পরবর্তী Computer-এ যেতে থাকে।

প্রত্যেক Computer Data Frame-এর **Destination MAC Address** পরীক্ষা করে।

- যদি MAC Address মিলে, তাহলে সেই Computer Data **Accept** করে।
- যদি না মিলে, তাহলে Data **পরবর্তী Computer-এর কাছে Forward** করে দেয়।

এভাবে Data Destination-এ পৌঁছানো পর্যন্ত Ring-এর মধ্যে ঘুরতে থাকে।

✔ Exam Ready Answer (English)

In a **Ring Topology**, data travels in a single direction (clockwise or counterclockwise) from one device to the next around the ring.

Each device receives the data frame and checks the **destination MAC address**.

- If the destination MAC address matches its own MAC address, the device accepts the data.
- Otherwise, it forwards the frame to the next device in the ring.

This process continues until the data reaches the intended destination.

Easy Example

Suppose **PC1** wants to send data to **PC3**.

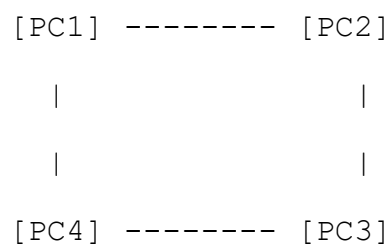
The data travels as follows:

PC1 → PC2 → PC3

- **PC2** checks the destination MAC address. Since it does not match, it forwards the data.
- **PC3** finds that the destination MAC address matches its own MAC address and accepts the data.

Figure

RING TOPOLOGY



Data Flow →

Advantages

- No Data Collision
- Equal Access

Disadvantages

- One Device Failure may affect the entire Network.

Example

Industrial Control Network

4. MESH TOPOLOGY

বাংলা Concept

Mesh Topology-তে প্রতিটি Computer অন্য সব Computer-এর সাথে **Direct Connection** রাখে।

তাই Data কোনো মধ্যবর্তী Device ছাড়াই সরাসরি Destination-এ যায়।

Destination Computer MAC Address মিলিয়ে Data গ্রহণ করে।

✔ Exam Ready Answer (English)

In **Mesh Topology**, every device has a direct connection with every other device.

Data travels directly to the destination without passing through intermediate devices.

The destination device checks the **destination MAC address** and accepts the data.

Easy Example

PC1 ————— PC4

PC1 directly sends data to PC4.

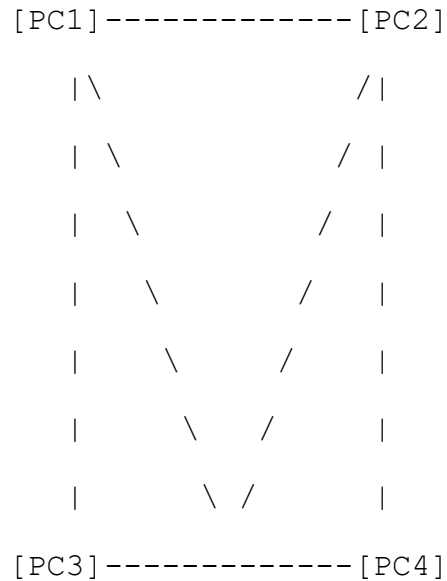
No Switch

No Hub

No Broadcast

Figure

MESH TOPOLOGY



★ এখানে প্রত্যেক PC অন্য সব PC-এর সাথে Connected।

Figure Explanation

- Every Device ↔ Every Device
- Multiple Communication Paths

Advantages

- Highly Reliable
- High Security

Disadvantages

- Expensive
- Complex

Example

5. TREE TOPOLOGY

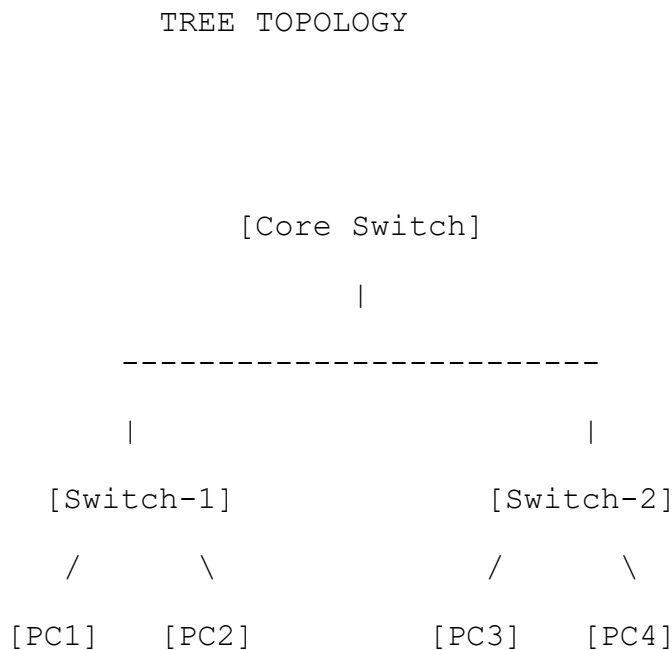
□ বাংলা Concept

Tree-এর মতো Layer বা Branch আকারে Connected থাকে।

✓ Exam Ready Answer

Tree Topology is a hierarchical topology that combines the characteristics of Bus and Star Topologies.

Figure



★ এটি একটি **Tree (গাছ)** এর মতো Root এবং Branch তৈরি করেছে।

Figure Explanation

- Core Switch → Root
- Switch-1, Switch-2 → Branch
- PCs → Leaf Nodes

Advantages

- Easy Expansion
- Suitable for Large Networks

Disadvantages

- Core Failure affects many devices.

Example

University Campus Network

6. HYBRID TOPOLOGY

□ বাংলা Concept

দুই বা ততোধিক Topology-এর সমন্বয়ে Hybrid Topology তৈরি হয়।

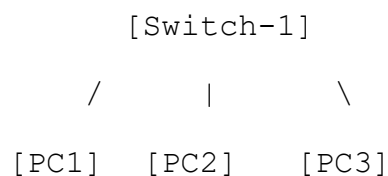
✓ Exam Ready Answer

Hybrid Topology is a combination of two or more different network topologies within a single network.

Figure

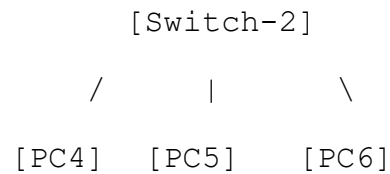
HYBRID TOPOLOGY

★ STAR NETWORK - 1



===== BACKBONE CABLE =====

★ STAR NETWORK - 2



Hybrid = Star + Bus

★ এখানে **দুইটি Star Network** একটি **Backbone Cable (Bus)** দিয়ে যুক্ত হয়েছে, তাই এটি **Hybrid Topology**।

Figure Explanation

- Switch-1 → First Star Network
- Switch-2 → Second Star Network
- Backbone Cable → Connects Both Star Networks

Advantages

- Flexible
- Scalable
- Reliable

Disadvantages

- Expensive
- Complex Design

Example

Large Banks, Corporate Offices

★ Exam Comparison (Very Important)

Topology	Shape	Main Device	Failure Effect	Used In
Bus	Straight Line	Backbone Cable	Backbone Failure → Whole Network Down	Small Office
Star	★ Star	Switch/Hub	Switch Failure → Whole Network Down	Banks
Ring	○ Circle	No Central Device	One Failure affects Ring	Industry
Mesh	Web/Net	No Central Device	Very Reliable	Military
Tree	♣ Tree	Core Switch	Core Failure affects Branches	University
Hybrid	Mixed	Multiple Devices	Depends on Design	Large Banks

✔ Example-1 (Class A)

Given:

IP Address = 10.50.20.100

Subnet Mask = 255.224.0.0

Step-1: Identify the Class

প্রথম Octet = 10

Class Range

Class Range

A 1–126

B 128–191

C 192–223

☞ 10 ⇒ Class A

Step-2: Default Subnet Mask

Class A-এর Default Mask

255.0.0.0

Binary

11111111.00000000.00000000.00000000

1-এর সংখ্যা

8

তাই Prefix

/8

Step-3: Given Mask

255.224.0.0

224-এর Binary

11100000

পুরো Binary

11111111.11100000.00000000.00000000

1-এর সংখ্যা

$$8 + 3 = 11$$

অতএব

$$255.224.0.0 = /11$$

Step-4: Borrowed Bits

Default ছিল

$$/8$$

Given হলো

$$/11$$

Borrowed Bits

$$11 - 8 = 3 \text{ Bits}$$

Step-5: Remaining Host Bits

Class A-এর Default Host Bits

$$24$$

Borrow করেছে

$$3$$

Remaining

$$24 - 3 = 21 \text{ Bits}$$

Step-6: Number of Subnets

Formula

$$2^{(\text{Borrowed Bits})}$$

$$2^3 = 8$$

✓ **8 Subnets**

Step-7: Number of Hosts

Formula

$$2^{(\text{Host Bits})} - 2$$

$$2^{21} - 2$$

$$= 2097152 - 2$$

$$= 2097150$$

✓ **2,097,150 Usable Hosts**

Step-8: Block Size

Formula

$$256 - 224$$

$$32$$

যেহেতু 224 দ্বিতীয় Octet-এ, তাই Block Size দ্বিতীয় Octet-এ হবে।

Network IDs

$$0$$

$$32$$

$$64$$

$$96$$

128

160

192

224

IP-এর দ্বিতীয় Octet

50

50 কোন Range-এ?

32-63

Step-9: Network Address

10.32.0.0

Broadcast Address

10.63.255.255

Valid Hosts

10.32.0.1

↓

10.63.255.254

Final Answer (Class A)

Item	Answer
Class	A

Item	Answer
Default Mask	255.0.0.0 (/8)
Given Mask	255.224.0.0 (/11)
Borrowed Bits	3 Bits
Host Bits	21 Bits
Subnets	8
Usable Hosts	2,097,150
Block Size	32
Network Address	10.32.0.0
Broadcast Address	10.63.255.255
Valid Hosts	10.32.0.1 – 10.63.255.254

✔ Example-2 (Class B)

Given:

IP Address = 172.16.35.100

Subnet Mask = 255.255.240.0

Step-1: Identify the Class

প্রথম Octet = 172

Range

128-191

👉 **Class B**

Step-2: Default Subnet Mask

255.255.0.0

Binary

11111111.11111111.00000000.00000000

1-এর সংখ্যা

$8 + 8 = 16$

তাই Prefix

/16

Step-3: Given Mask

255.255.240.0

240-এর Binary

11110000

পুরো Binary

11111111.11111111.11110000.00000000

1-এর সংখ্যা

$8 + 8 + 4 = 20$

অতএব

255.255.240.0 = /20

Step-4: Borrowed Bits

Default

/16

Given

/20

Borrowed

$20 - 16 = 4 \text{ Bits}$

Step-5: Remaining Host Bits

Class B-এর Host Bits

16

Borrow করেছে

4

Remaining

$16 - 4 = 12 \text{ Bits}$

Step-6: Number of Subnets

Formula

$2^{(\text{Borrowed Bits})}$

$2^4 = 16$

✓ **16 Subnets**

Step-7: Number of Hosts

Formula

$$2^{(\text{Host Bits})} - 2$$

$$2^{12} - 2$$

$$= 4096 - 2$$

$$= 4094$$

✓ **4094 Usable Hosts**

Step-8: Block Size

Formula

$$256 - 240$$

$$16$$

Network IDs

0

16

32

48

64

80

96

112

IP-এর তৃতীয় Octet

35

35 কোন Range-এ?

32-47

Step-9: Network Address

172.16.32.0

Broadcast Address

172.16.47.255

Valid Hosts

172.16.32.1

↓

172.16.47.254

Final Answer (Class B)

Item	Answer
Class	B
Default Mask	255.255.0.0 (/16)
Given Mask	255.255.240.0 (/20)
Borrowed Bits	4 Bits

Item	Answer
Host Bits	12 Bits
Subnets	16
Usable Hosts	4094
Block Size	16
Network Address	172.16.32.0
Broadcast Address	172.16.47.255
Valid Hosts	172.16.32.1 – 172.16.47.254

★ তিনটি Class-এর একসাথে Comparison (সবচেয়ে গুরুত্বপূর্ণ)

Item	Class A	Class B	Class C
IP Example	10.50.20.100	172.16.35.100	192.168.10.25
Default Mask	255.0.0.0	255.255.0.0	255.255.255.0
Default Prefix	/8	/16	/24
Default Host Bits	24	16	8
Given Prefix	/11	/20	/27

Item	Class A	Class B	Class C
Borrowed Bits	3	4	3
Remaining Host Bits	21	12	5
Subnets	8	16	8
Usable Hosts	2,097,150	4,094	30

Example (Class C)

Given:

IP Address = 192.168.10.25

Subnet Mask = 255.255.255.224

Step-1: Identify the Class

প্রথম Octet = 192

Class Range:

Class Range

A 1–126

B 128–191

C 192–223

☞ 192 ⇒ Class C

Step-2: Write the Default Subnet Mask

Class C-এর Default Mask:

255.255.255.0

Binary:

11111111.11111111.11111111.00000000

1-এর সংখ্যা

$8+8+8=24$

তাই Prefix

/24

Step-3: Convert Given Mask into Binary

Given Mask

255.255.255.224

Binary

255 = 11111111

255 = 11111111

255 = 11111111

224 = 11100000

সবগুলো একসাথে

11111111.11111111.11111111.11100000

1-এর সংখ্যা

$$8+8+8+3=27$$

অতএব

$$255.255.255.224 = /27$$

Step-4: Borrowed Bits

Default ছিল

$$/24$$

এখন

$$/27$$

Borrow করেছে

$$27-24=3 \text{ Bits}$$

👉 Host Portion থেকে 3 Bits Borrow করা হয়েছে।

Step-5: Remaining Host Bits

Class C-তে Host Bits ছিল

$$8 \text{ Bits}$$

Borrow করেছে

$$3 \text{ Bits}$$

Remaining

$$8-3=5 \text{ Bits}$$

Step-6: Number of Subnets

Formula

$2^{\text{(Borrowed Bits)}}$

$2^3=8$

✓ **8 Subnets**

Step-7: Number of Hosts

Formula

$2^{\text{(Host Bits)}} - 2$

$2^5 - 2$

$32 - 2$

30

✓ **30 Usable Hosts**

? এখানে "-2" কেন?

এটি 2 Bits না, 2টি Host Address।

অনেকে ভুল করে বলে "২টা Bit Minus করি"।

✗ এটা ভুল।

সঠিক কথা হলো—

আমরা ২টি Address ব্যবহার করতে পারি না।

কারণ—

প্রথম Address = Network Address

উদাহরণ

192.168.10.0

এটি পুরো Network-কে বোঝায়।

কোনো Computer-কে দেওয়া যায় না।

শেষ Address = Broadcast Address

192.168.10.31

এটি পুরো Network-এ Message পাঠানোর জন্য Reserved।

এটিও কোনো Host-কে দেওয়া যায় না।

তাই

মোট Address

32

কিন্তু

1 Network Address

+

1 Broadcast Address

Reserved।

তাই

$32 - 2 = 30$

☞ **Minus 2 মানে 2টি Address, Bit নয়।**

Step-8: Block Size

Formula

$256 - 224$

32

Block Size

32

Network Numbers হবে

0

32

64

96

128

160

192

224

Step-9: Network Address

শেষ Octet

25

25 কোন Range-এ?

0-----31

অতএব

Network

192.168.10.0

Broadcast

192.168.10.31

Valid Hosts

192.168.10.1

↓

192.168.10.30

Final Answer

Item	Answer
Class	C
Default Mask	255.255.255.0 (/24)
Given Mask	255.255.255.224 (/27)

Item	Answer
Borrowed Bits	3 Bits
Remaining Host Bits	5 Bits
Number of Subnets	8
Number of Usable Hosts	30
Block Size	32
Network Address	192.168.10.0
Broadcast Address	192.168.10.31
Valid Host Range	192.168.10.1 – 192.168.10.30

OSI (Open Systems Interconnection) Model

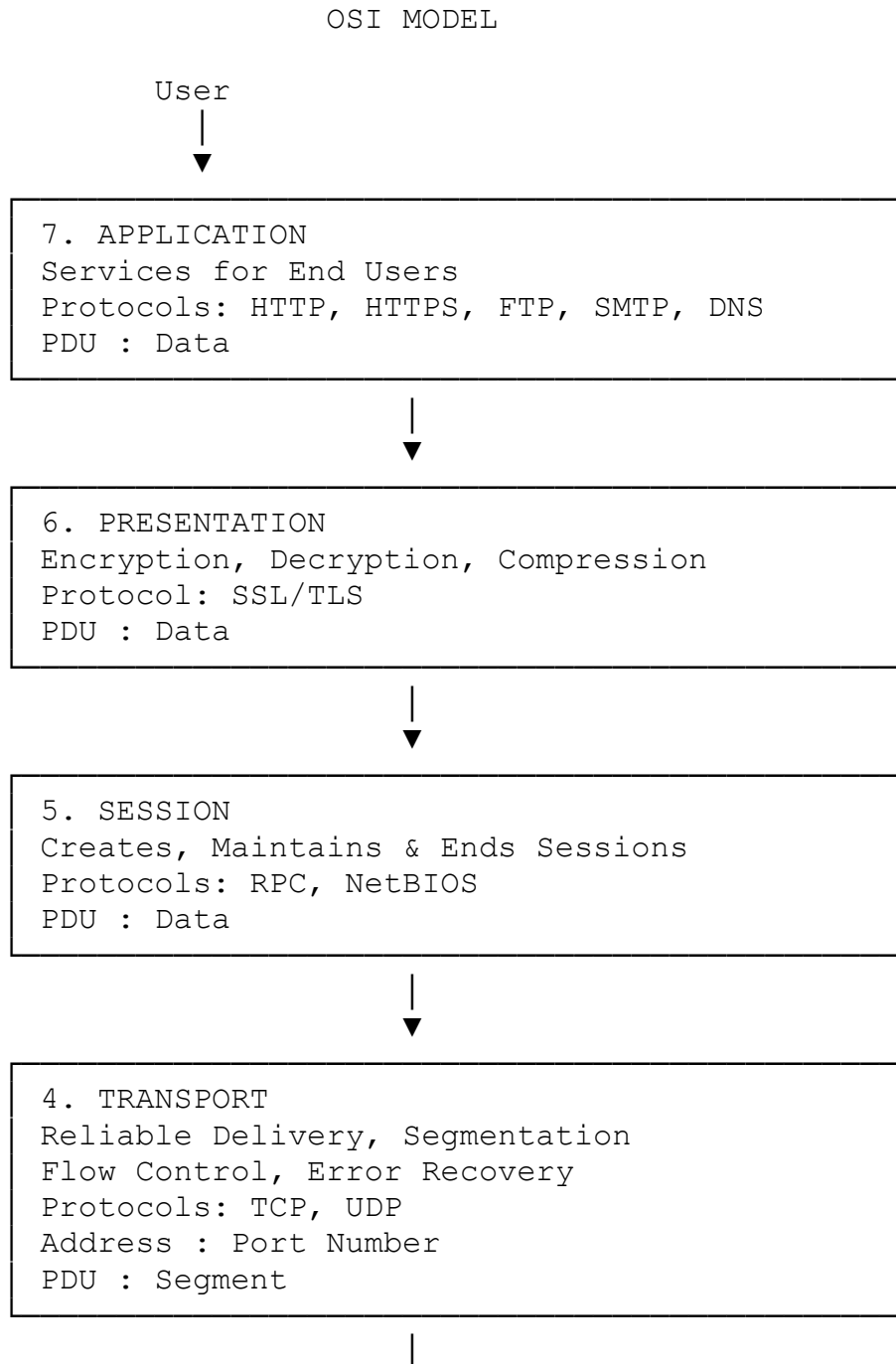
□ Concept (বাংলা)

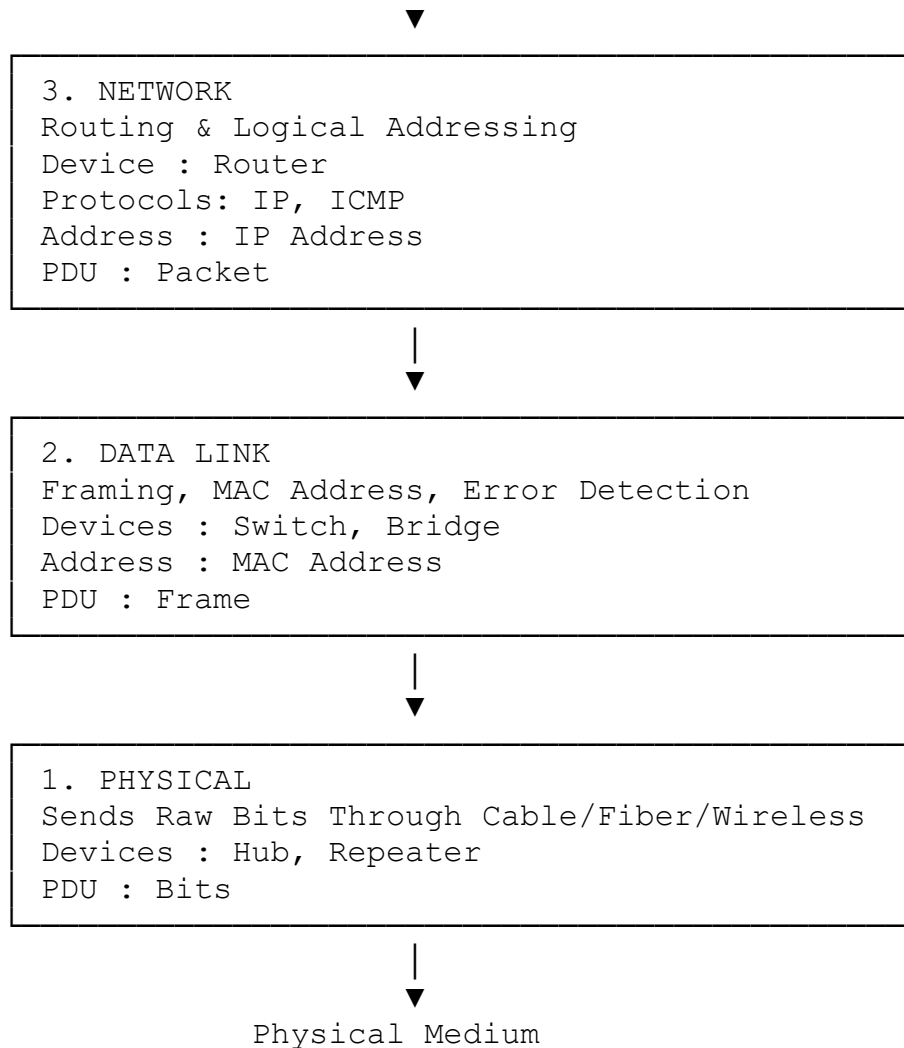
OSI Model হলো ISO কর্তৃক তৈরি একটি **7-layer Reference Model**, যা দেখায় একটি Computer থেকে অন্য Computer-এ Data কীভাবে ধাপে ধাপে আদান-প্রদান হয়। প্রতিটি Layer-এর নির্দিষ্ট কাজ রয়েছে এবং প্রতিটি Layer উপরের Layer-কে Service প্রদান করে।

✓ Exam Ready Answer (English)

The OSI (Open Systems Interconnection) Model is a seven-layer reference model developed by the International Organization for Standardization (ISO). It standardizes network communication by dividing the communication process into seven logical layers, where each layer performs a specific function and provides services to the layer above it.

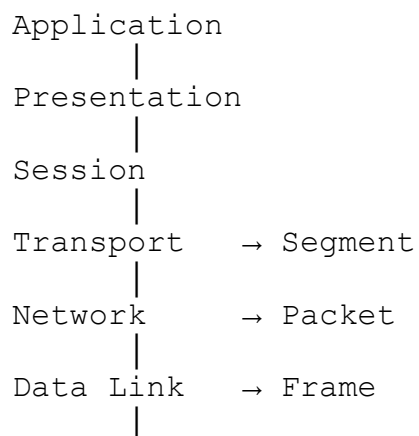
OSI Model (Graphical View)





Data Flow (Very Important)

Sender Side (Encapsulation)



Physical → Bits
|
Transmission Media

Receiver Side (Decapsulation)

Transmission Media
|
Physical → Bits
|
Data Link → Frame
|
Network → Packet
|
Transport → Segment
|
Session
|
Presentation
|
Application

PDU (Protocol Data Unit)

Layer	PDU
Application	Data
Presentation	Data
Session	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

Address Used

Layer	Address
Transport	Port Number
Network	IP Address
Data Link	MAC Address

Networking Devices

Device	Layer
Hub	Physical
Repeater	Physical
Switch	Data Link
Bridge	Data Link
Router	Network
Gateway	Application / Multiple Layers

Easy Banking Example

Suppose a customer transfers **₹50,000** using the **National Bank Mobile App**.

Application
User clicks Transfer

↓

Presentation
Encrypts the data (SSL/TLS)

↓

Session
Maintains the connection

↓

Transport
TCP ensures reliable delivery



Network
Uses IP Address to reach the Bank Server



Data Link
Uses MAC Address within the local network



Physical
Sends Bits through Wi-Fi/Fiber Cable

Easy Banking Example

Suppose a customer transfers **₹10,000** using the **Pubali Bank Mobile App**.

- **Application Layer** → User requests the transfer.
- **Presentation Layer** → Encrypts the data using SSL/TLS.
- **Session Layer** → Establishes and manages the session.
- **Transport Layer** → Uses TCP to ensure reliable delivery.
- **Network Layer** → Uses the destination IP address to route the packet.
- **Data Link Layer** → Uses the destination MAC address to deliver the frame within the local network.
- **Physical Layer** → Sends the data as bits through Wi-Fi, Ethernet, or Fiber.

Advantages of OSI Model

- Standardizes network communication.
- Simplifies network design and troubleshooting.
- Provides modular architecture.
- Ensures interoperability between different vendors.
- Makes maintenance and upgrades easier.

Disadvantages of OSI Model

- More theoretical than practical.
- Complex compared to the TCP/IP model.
- Some layers have overlapping functions.